Panic Disorder Detection Project

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COSC 4610: Data Mining

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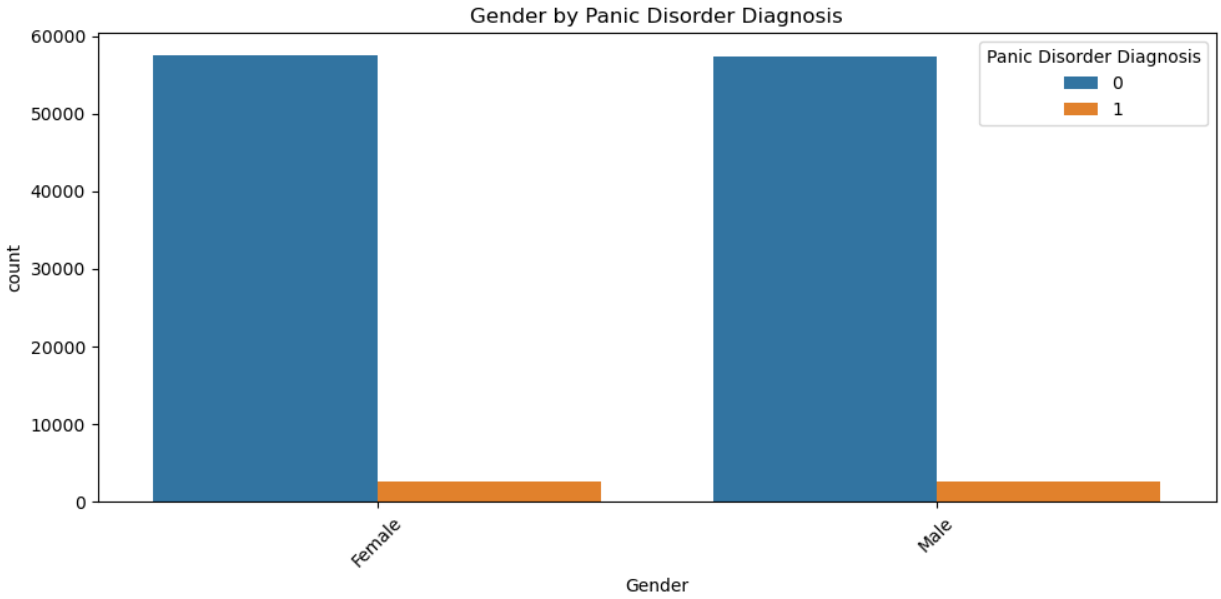
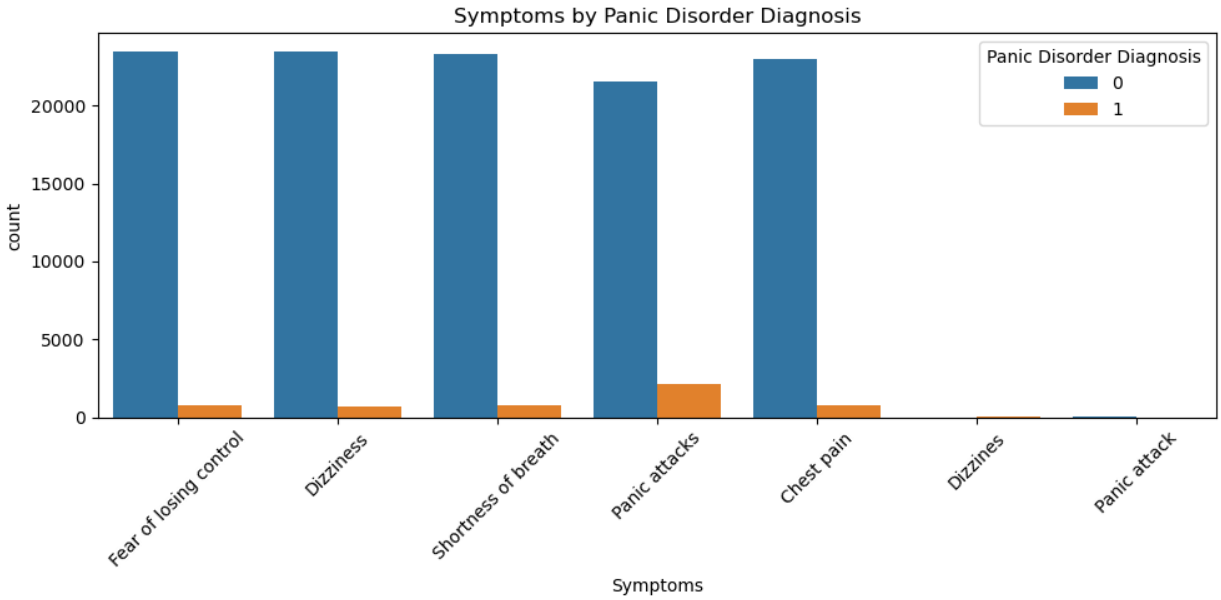
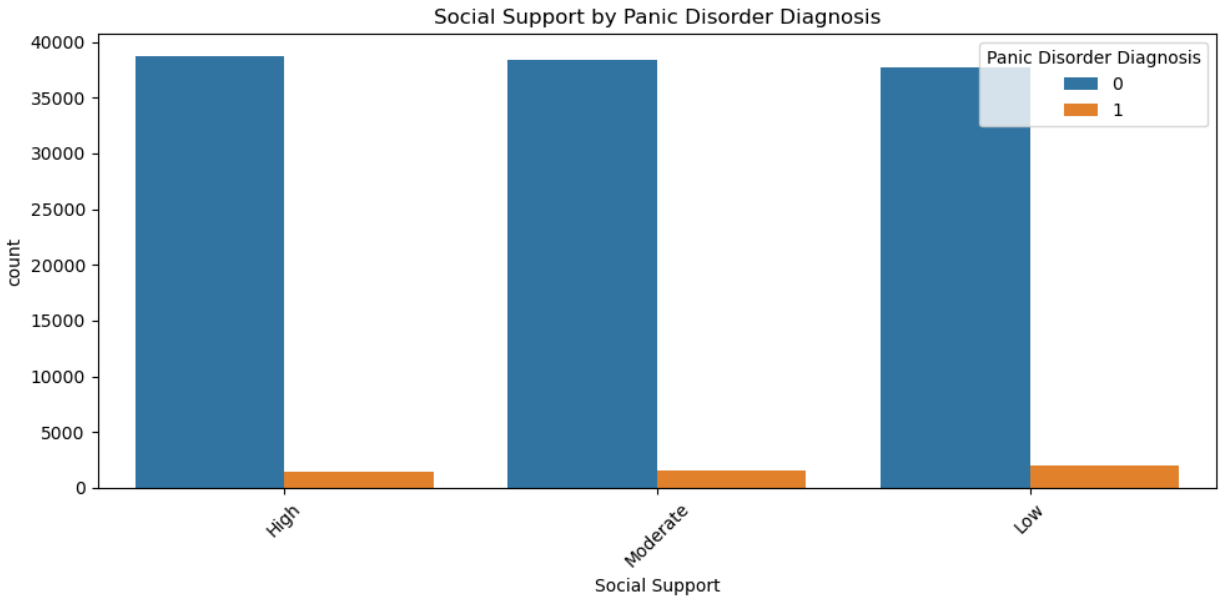
Step 1: Data Exploration and Understanding

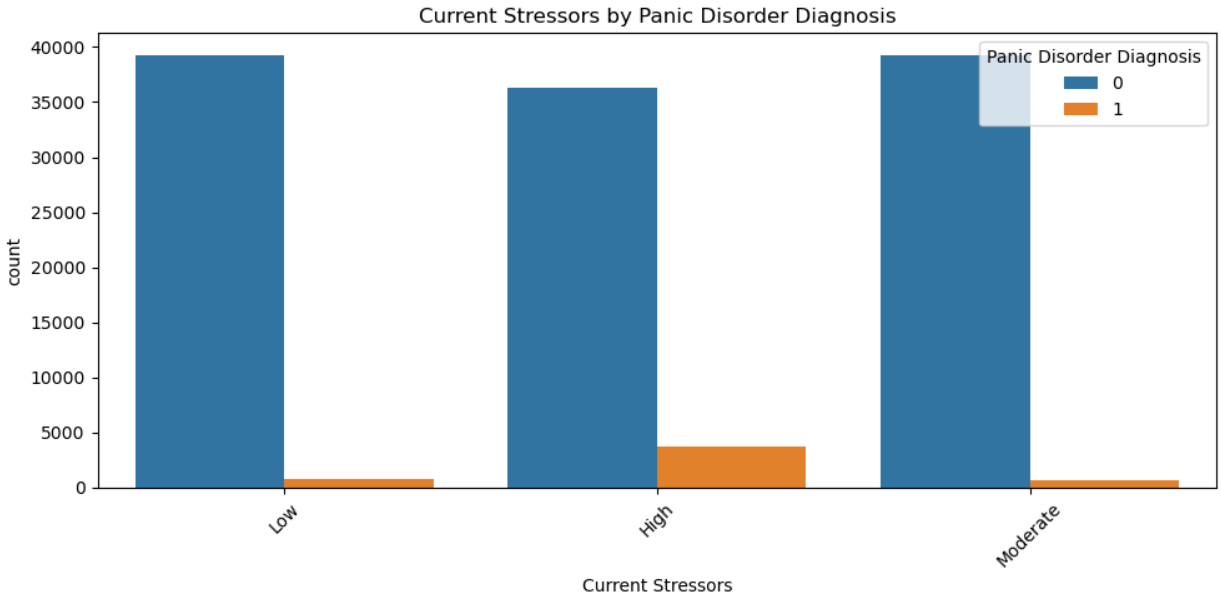
Observations about the dataset: there are 120,000 rows and 17 columns. There are 30,224 missing values in the 'Medical History' field, 29,896 missing values in the 'Psychiatric History' field, and 40,078 missing values in the 'Substance Use' field. There are 19,601 duplicate rows in the dataset.

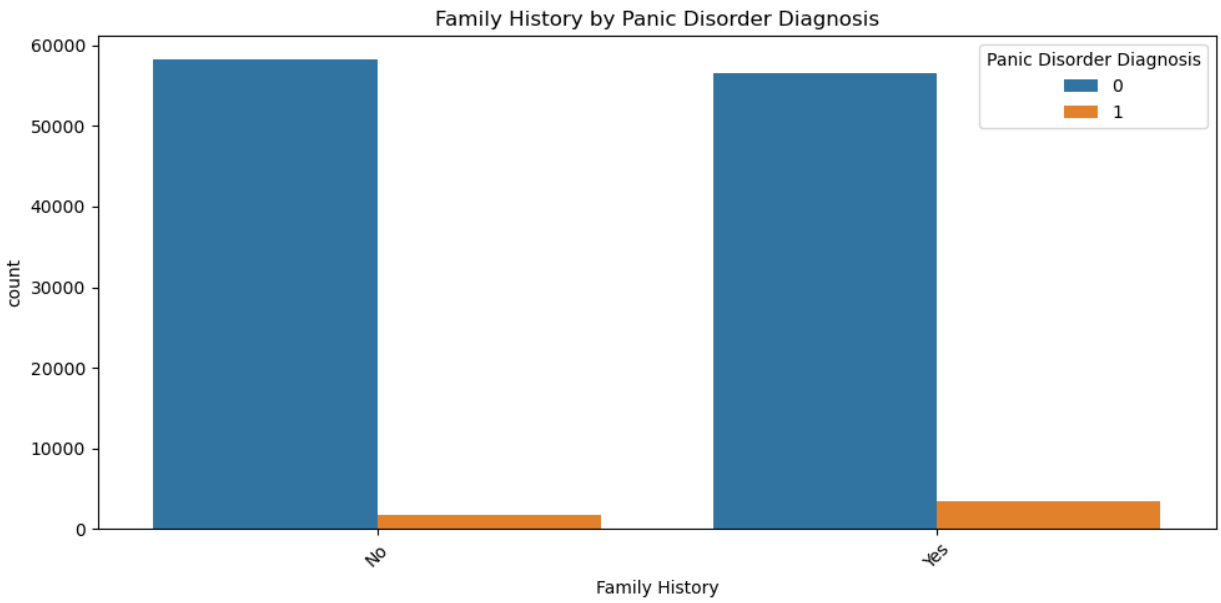
These are the visual representations of categorical variables versus diagnosis, including gender, symptoms, social support, current stressors, and family history:



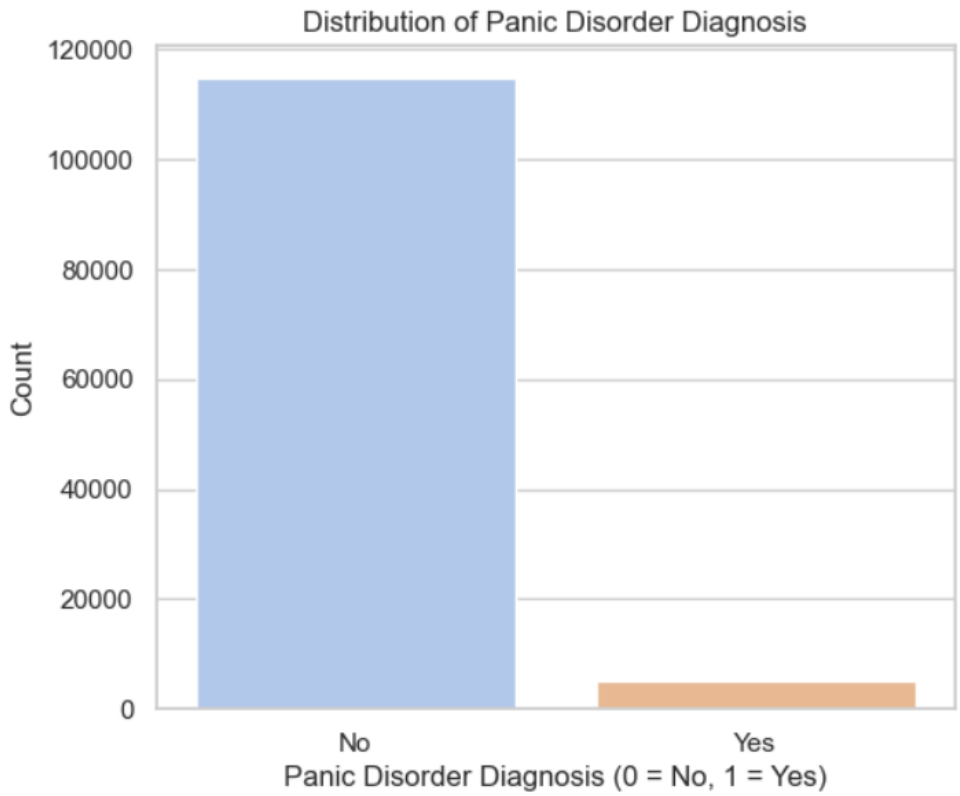



Social Support by Panic Disorder Diagnosis
Panic Disorder Diagnosis
0
1
count
40000
35000
30000
25000
20000
15000
10000
5000
0
High
Moderate
Low
Social Support


Current Stressors by Panic Disorder Diagnosis
Panic Disorder Diagnosis
0
1
count
40000
35000
30000
25000
20000
15000
10000
5000
0
Low
High
Moderate
Current Stressors

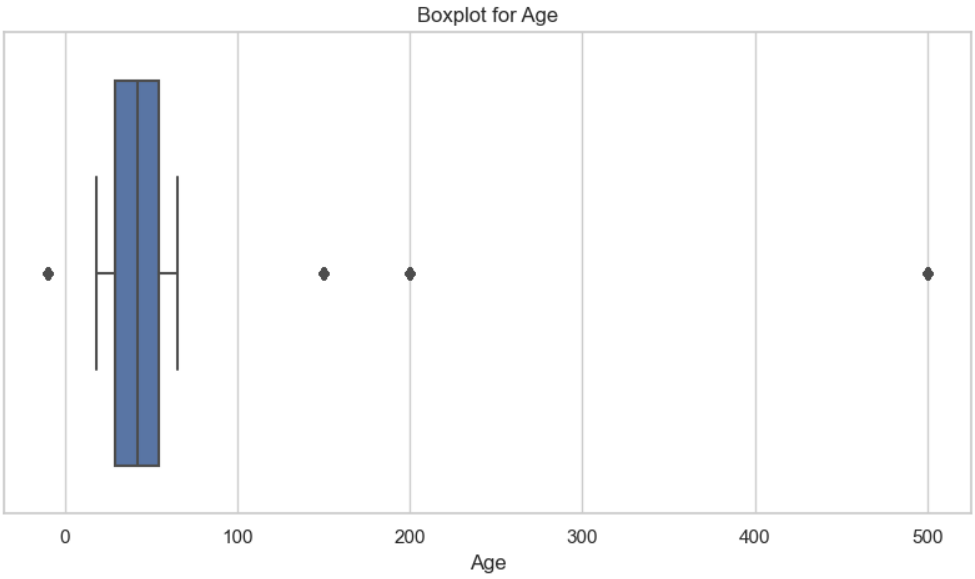


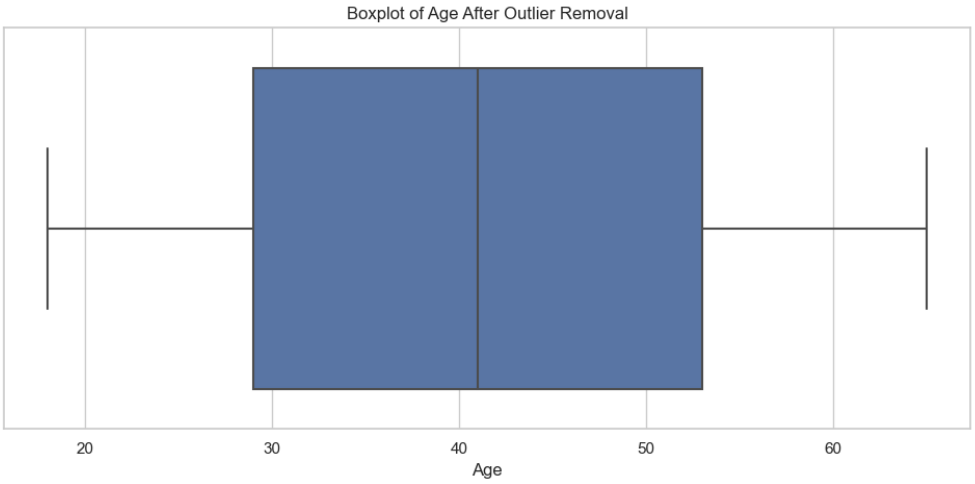
There exists class imbalance in the target variable: 114,858 had no diagnosis and 5,142 had a diagnosis. This is the bar chart:



Step 2: Data Cleaning and Transformation

When handling missing values, I dropped the empty 'Substance Use' column since it had the largest number of missing values (greater than 30% of the dataset was missing this information) and whether the individual used a substance or not is likely not very informative when it comes to panic disorder detection. I also dropped the 'Participant ID' column because as I was working through the project, I noticed that the Random Forest model listed it as an important feature, so I wanted to eliminate this bias. To handle missing categorical values, I used mode so that the values would represent the overall trends in the data. Next, I dropped duplicate values and ran a check to ensure that the label proportions hadn't changed. Then I converted the categorical values to the proper string format and checked for date related fields. Finally, I used boxplots to look for outliers and noticed some inconsistencies with the age field, so I used IQR to remove the outliers.





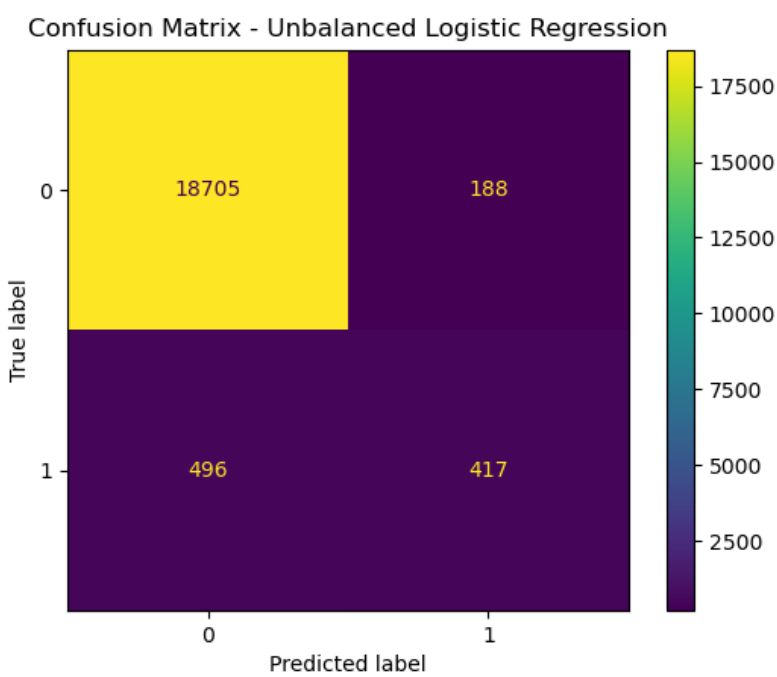
Step 3: Handling Class Imbalance

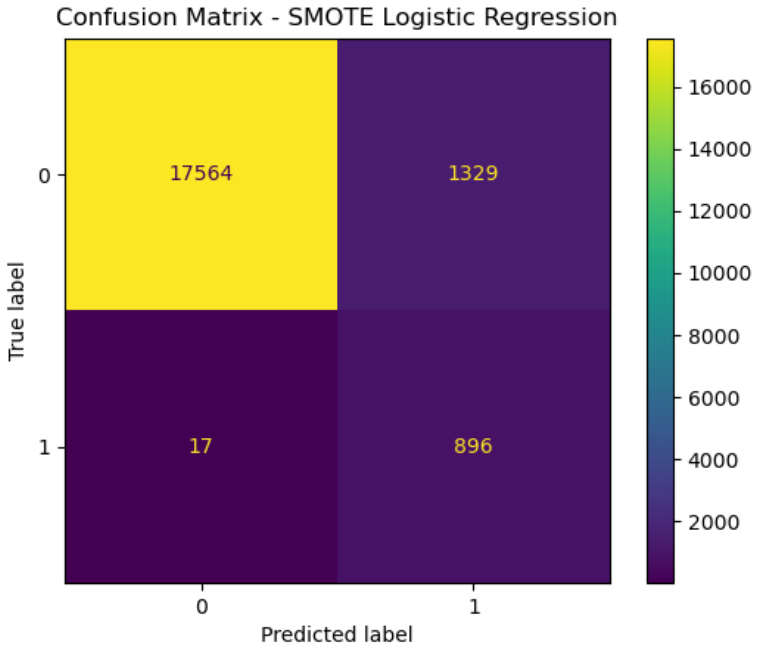
Prior to class balancing, the class imbalance ratio was 94,771 : 4,255. I then applied SMOTE oversampling and random oversampling of the minority class. After applying the class imbalance handling techniques, each class had 75,878 samples for SMOTE and random oversampling of the minority class. I then compared the performance on a Linear Regression model (see table below).

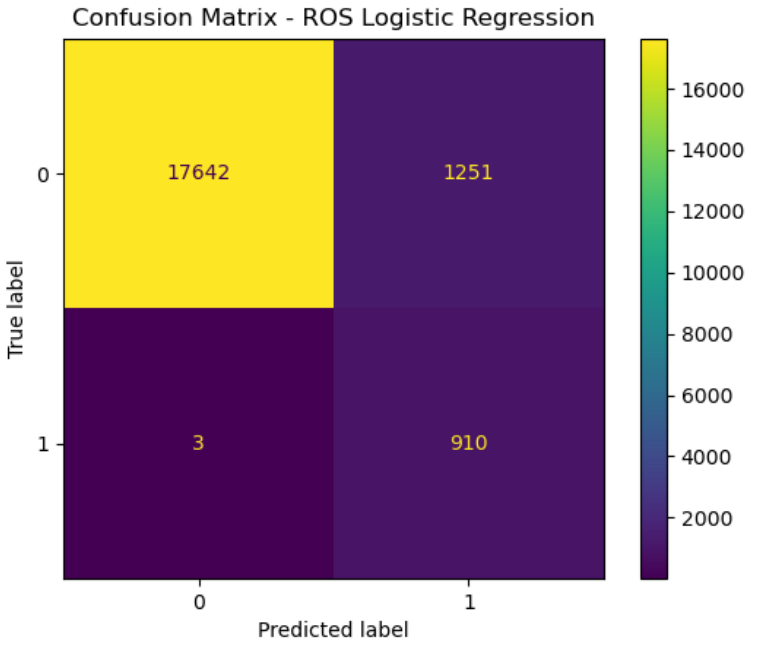
Performance comparison (Linear Regression model):

Metric	No balancing	With SMOTE	With random oversampling
Recall (Class 1)	0.46	0.98	1.00
Precision (Class 1)	0.68	0.40	0.42
F1 Score (Class 1)	0.55	0.57	0.59
Accuracy	0.97	0.93	0.94

Confusion matrices:







Looking at the no balancing case first, it has low recall and many of the true positives were missed. The precision was also relatively low, so it made a lot of false positive predictions. The F1 score indicates a low balance of precision and recall, indicating that it was not very effective at detecting the minority class. Overall, the no balancing case is biased toward the majority class,

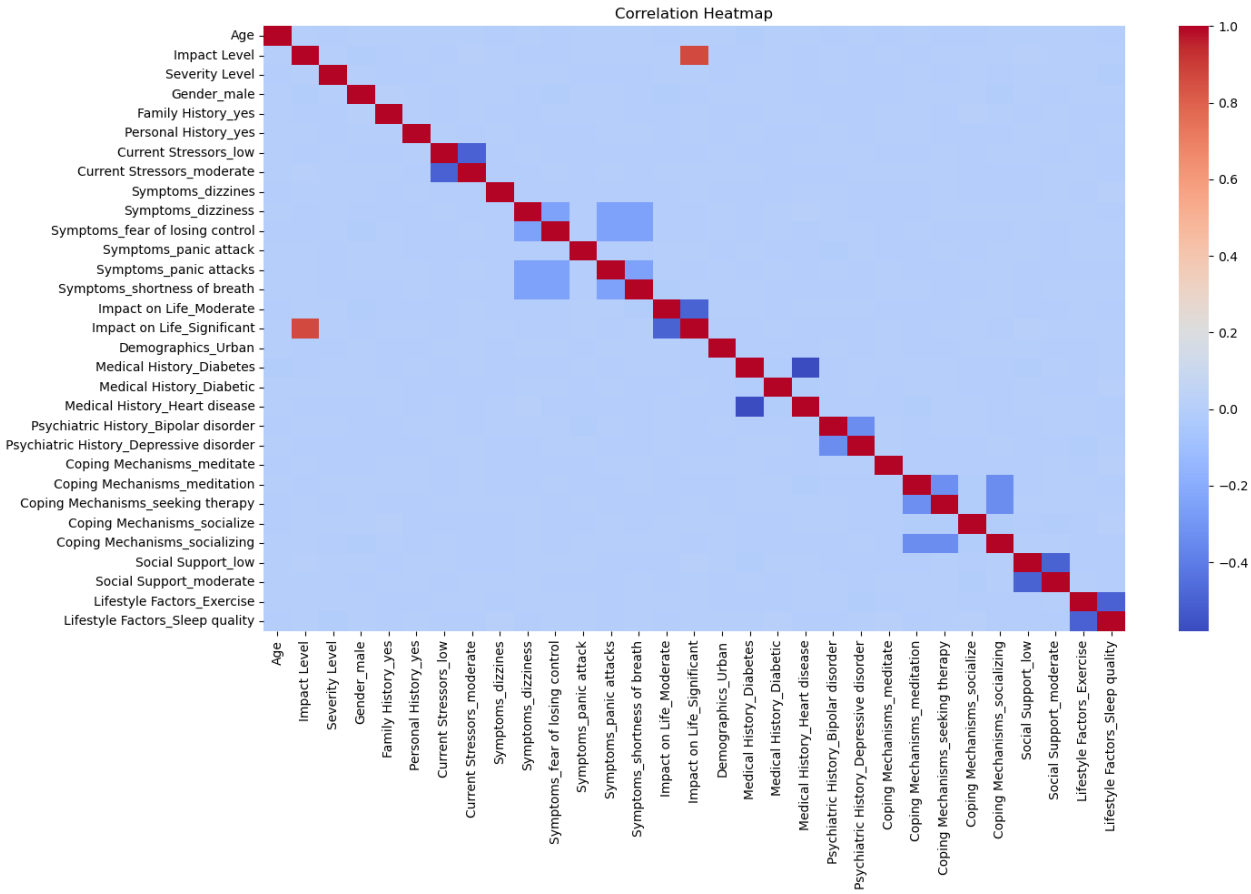
missing many actual positive cases which is not good for a medical diagnosis scenario. With SMOTE balancing, the precision dropped quite low (0.4), indicating that the model is identifying many positive cases, most of which are false positives. The recall was quite high though (0.98), so after SMOTE balancing, the model was able to accurately identify many of the positive cases. The F1 score for SMOTE is low, indicating that precision can still be improved, but in medical scenarios, better recall is usually preferred. Finally, with random oversampling of the minority class, it had perfect recall meaning all the positive cases were detected, a vast improvement compared to no balancing. The precision is slightly better than SMOTE, but it is still relatively low and there are still many false positives being detected. It did have the highest F1 score, however, and while precision could be improved, it had the best balance of all 3 techniques.

Class balancing is important in medical diagnosis models because it is important to help the patient get treatment. If false negatives occur, treatment can be delayed, and the patient's health may decline. Class balancing ensures that the model gives fair attention to minority classes, and as the data above suggests, class balancing improves a model's ability to detect true cases even when they are rare in the dataset.

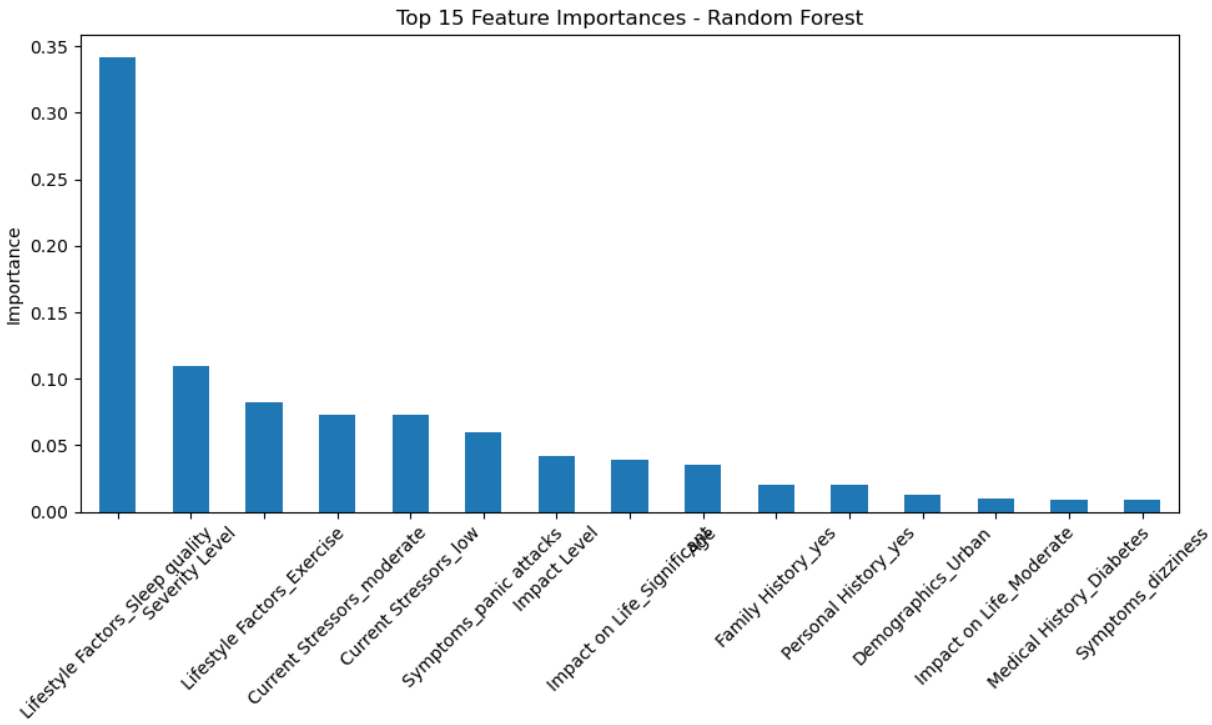
Step 4: Feature Engineering

I used one-hot encoding on categorical variables to help the models better interpret the data. One-hot encoding helps to avoid unintended ordinal relationships between categories, and each column had at most 3 categories, so it did not add too many new columns to work with. I then used z-score scaling to standardize numeric features on the same scale. For models like Logistic Regression and kNN, which rely on distances between the data points, features with different scales can cause poor model performance. Z-score scaling also ensures that features with larger ranges would not disproportionately influence the model. Finally, I mapped the Impact Level and Severity Level to represent the intensity of each feature. Both features were ordinal, so creating new features for them allowed the machine learning models to better interpret the data.

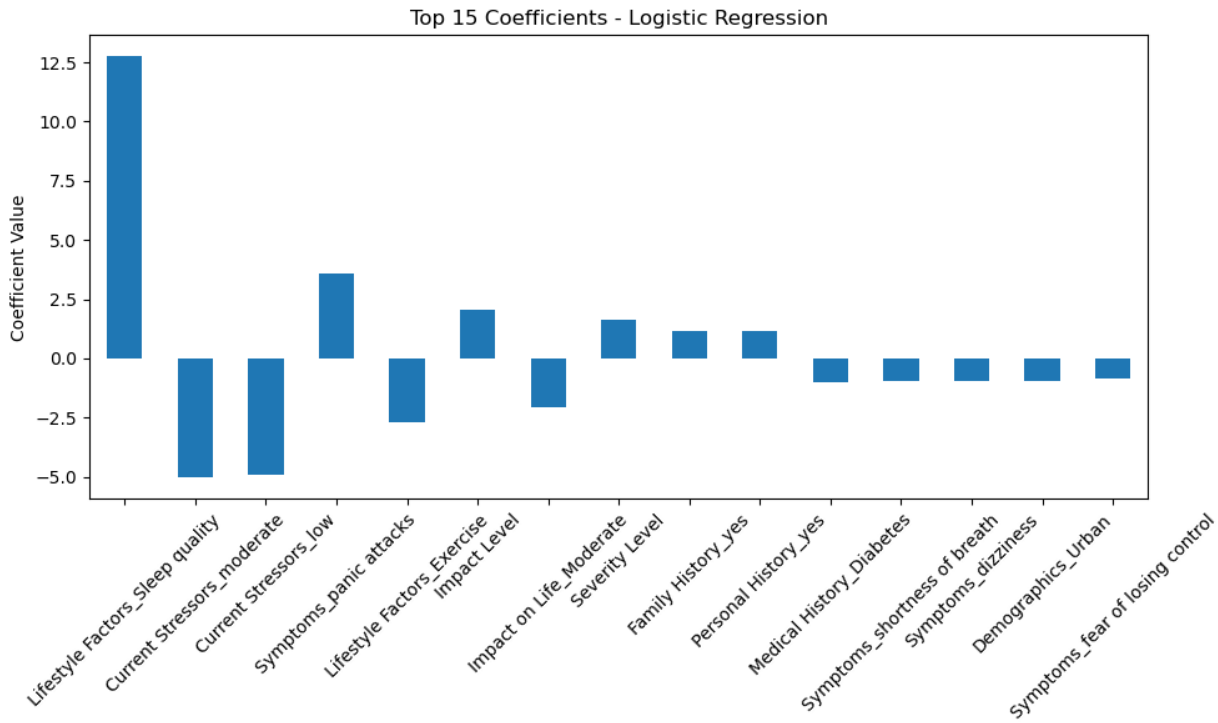
Step 5: Feature Importance Analysis



The heatmap above seems to show a moderate negative correlation between age and severity, possibly indicating that older individuals experience less severe symptoms.



The features importances from the Random Forest model give insight into which features contribute to predictions. Sleep quality has the highest importance score, suggesting that a person's quality of sleep plays a significant role in predicting whether someone has a panic disorder. Other features like exercise and current stress levels also contribute to predictions, which makes sense because exercise can overall impact mental health and stress management, and people experiencing higher amounts of stress are probably more likely to experience symptoms of panic disorder.

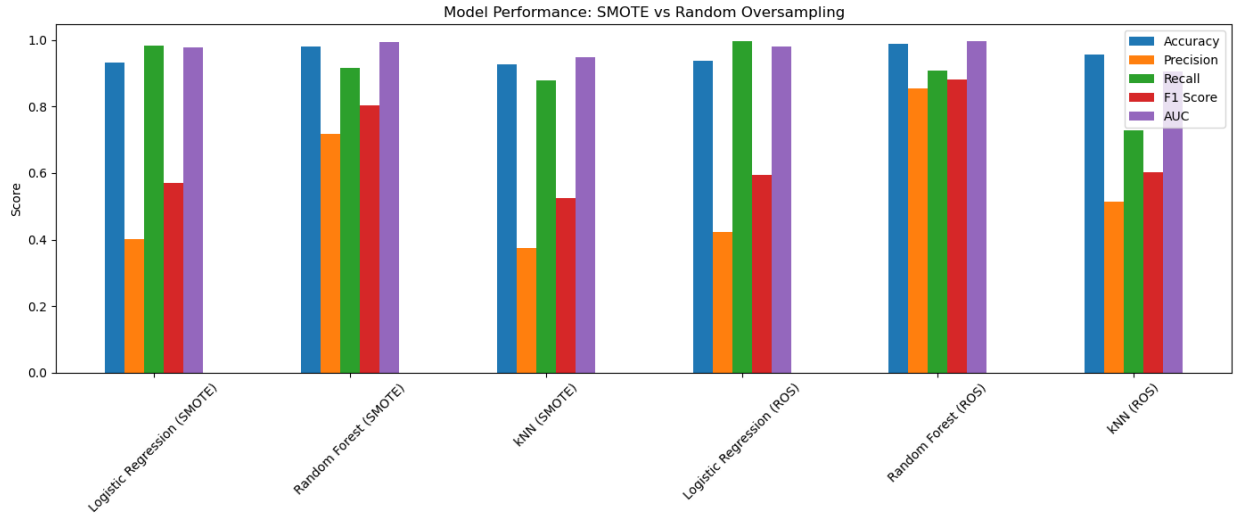


Looking at the Logistic Regression coefficients, sleep quality has the highest positive coefficient, indicating again that sleep quality is one of the highest contributors to predictions. Also, moderate stressors have a relatively high positive coefficient, again indicating that stressors will impact predictions.

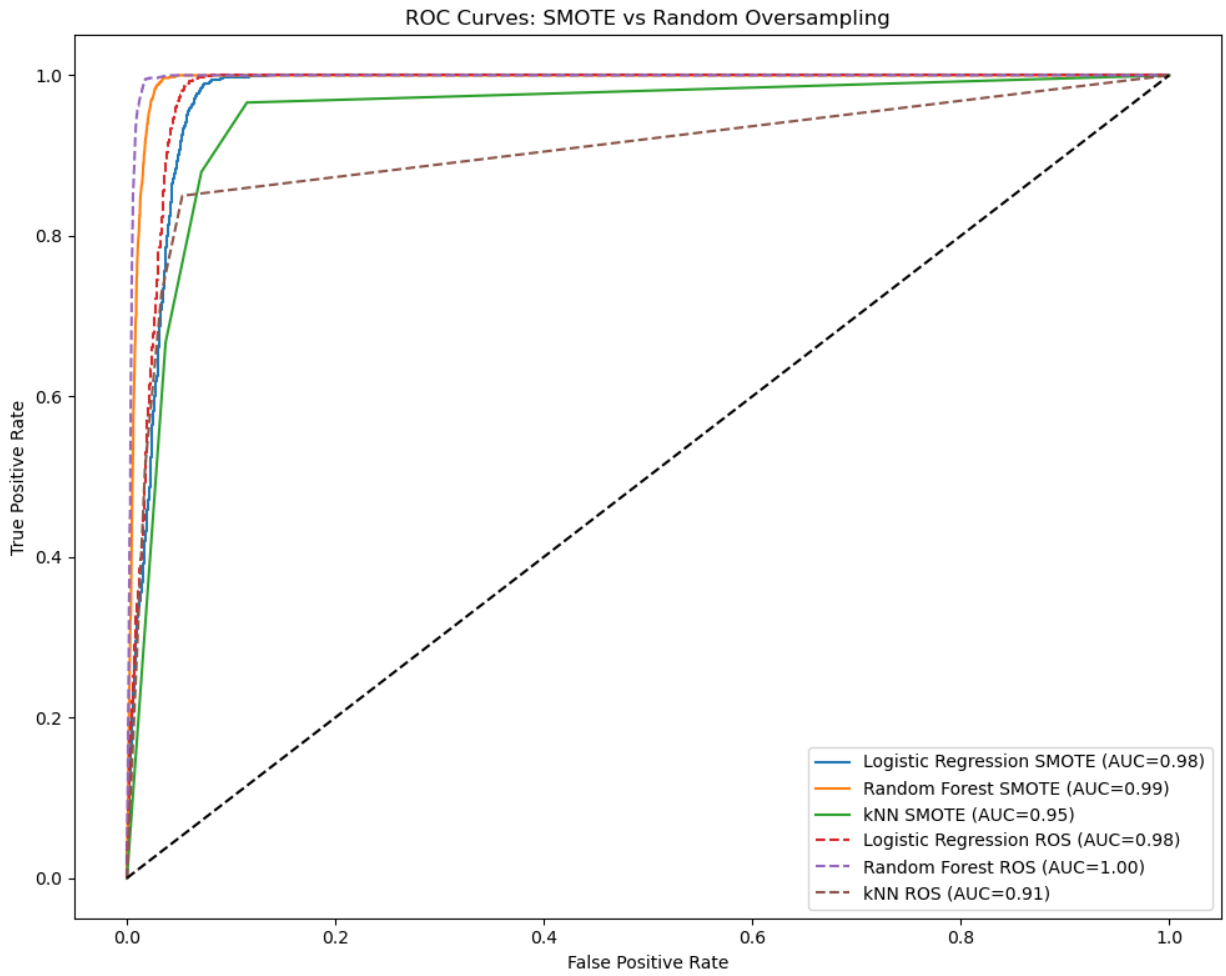
Step 6: Model Training and Evaluation

Performance metrics overview:

	Accuracy	Precision	Recall	F1 Score	AUC
Logistic Regression (SMOTE)	0.931990	0.402603	0.982475	0.571156	0.976127
Random Forest (SMOTE)	0.979400	0.716738	0.914567	0.803657	0.992789
kNN (SMOTE)	0.926386	0.373315	0.879518	0.524151	0.949127
Logistic Regression (ROS)	0.936989	0.422202	0.995619	0.592955	0.980967
Random Forest (ROS)	0.988539	0.852881	0.907996	0.879576	0.996275
kNN (ROS)	0.955872	0.515105	0.728368	0.603448	0.905814



From these performance metrics, Random Forest (both SMOTE and random oversampling of minority class) shows top overall performance. Of the three, Logistic Regression has the best recall, which is critical for medical diagnoses. kNN lags in precision and F1 scores, and seems to be less reliable in this scenario. Overall, Random Forest combined with random oversampling of the minority class has the best performance, combining relatively high precision and recall.



Looking at the ROC curves, it is clear that all of the models benefit from class balancing. Additionally, the data cleaning and feature selection had positive impacts on the ROC curve, resulting in higher and cleaner curves. kNN with random oversampling of the minority class has a slightly flatter curve and AUC, indicating that it is less reliable under random oversampling. Random Forest with ROC has a perfect AUC score, indicating that it can distinguish between positive and negative cases very well, which is good for medical scenarios.

Step 7: Ethical Considerations

Bias in mental health predictions can arise when models are trained on biased data, leading to potential unfair outcomes for certain groups. For example, if a model was trained on data from one demographic like gender, age, or ethnic group, it will not likely be applicable to other groups. This can lead to misdiagnosis or underdiagnosis for unrepresented groups. Ensuring that data is representative of a diverse range of people is important for models to be more accurate and equitable.

Using AI for mental health assessments can raise several ethical questions and concerns. One issue is patient privacy, as mental health data can be very sensitive. Additionally, AI might be

misused to diagnose or treat patients, when it is important for medical professionals to have a role in assessing the person. Every individual is unique and has unique experiences, and AI might not be able to capture all that information. Also, transparency is important when it comes to AI, especially in the medical field, because medical providers should understand how AI models make predictions and be able to use AI responsibly.